10. Transformation-Based POS Tagging

Aim: To implement transformation-based tagging using simple rules to modify initial POS tags.

Procedure:

1. Assign initial POS tags to words.
2. Define transformation rules.
3. Check each word against the rules.
4. Update the tags when conditions match.
5. Display the final tags.

Code:

```
words = ["The", "dog", "is", "playing"]

tags = ["NN", "NN", "NN", "NN"]

for i in range(len(words)):

   if words[i] == "The":

      tags[i] = "DT"

   elif words[i] == "is":

      tags[i] = "VBZ"

   elif words[i].endswith("ing"):

      tags[i] = "VBG"

for word, tag in zip(words, tags):

   print(word, ":", tag)
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/10.py"
The : DT
dog : NN
is : VBZ
playing : VBG
```

Result: Transformation rules were successfully applied to assign POS tags to words.